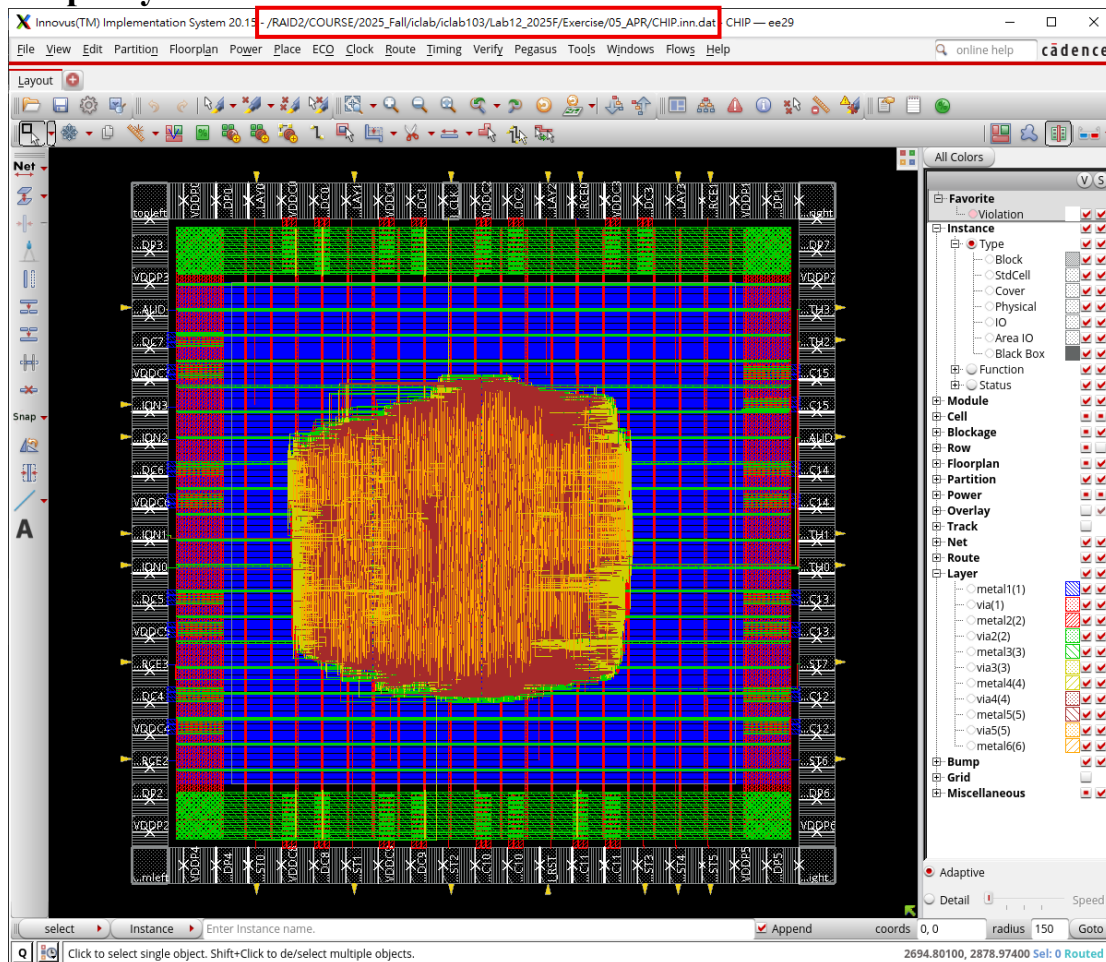


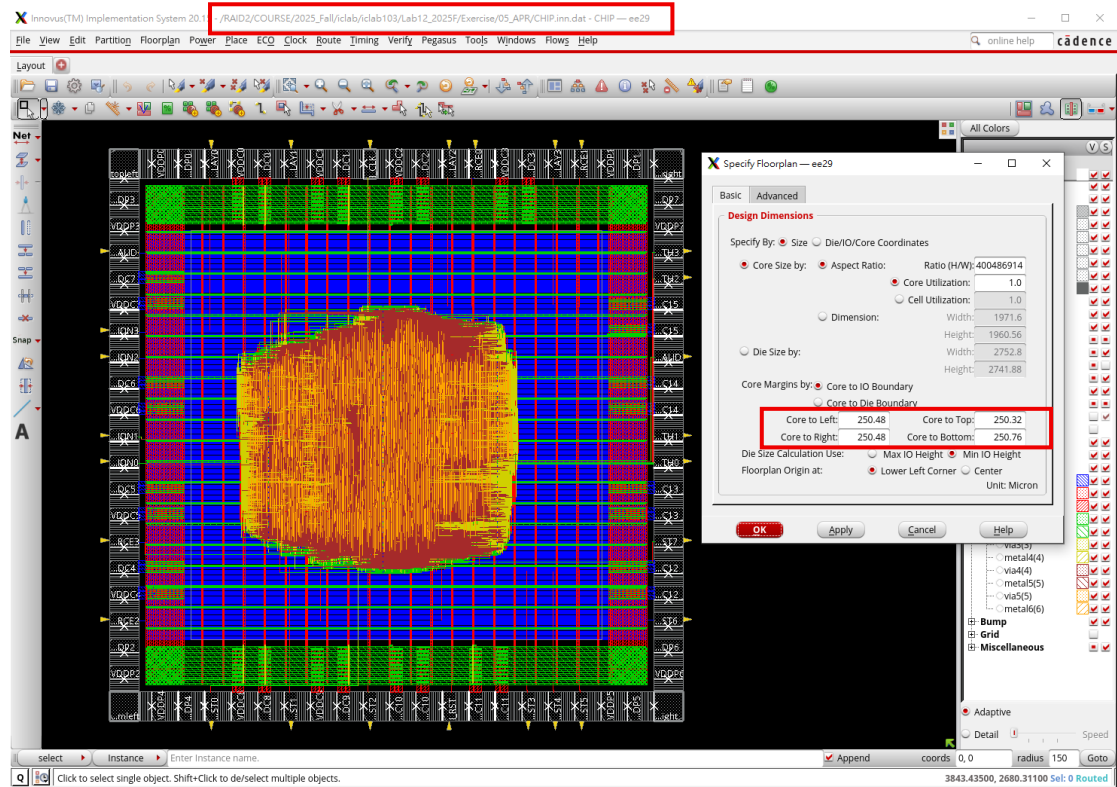
Report

314510200 林佳葦

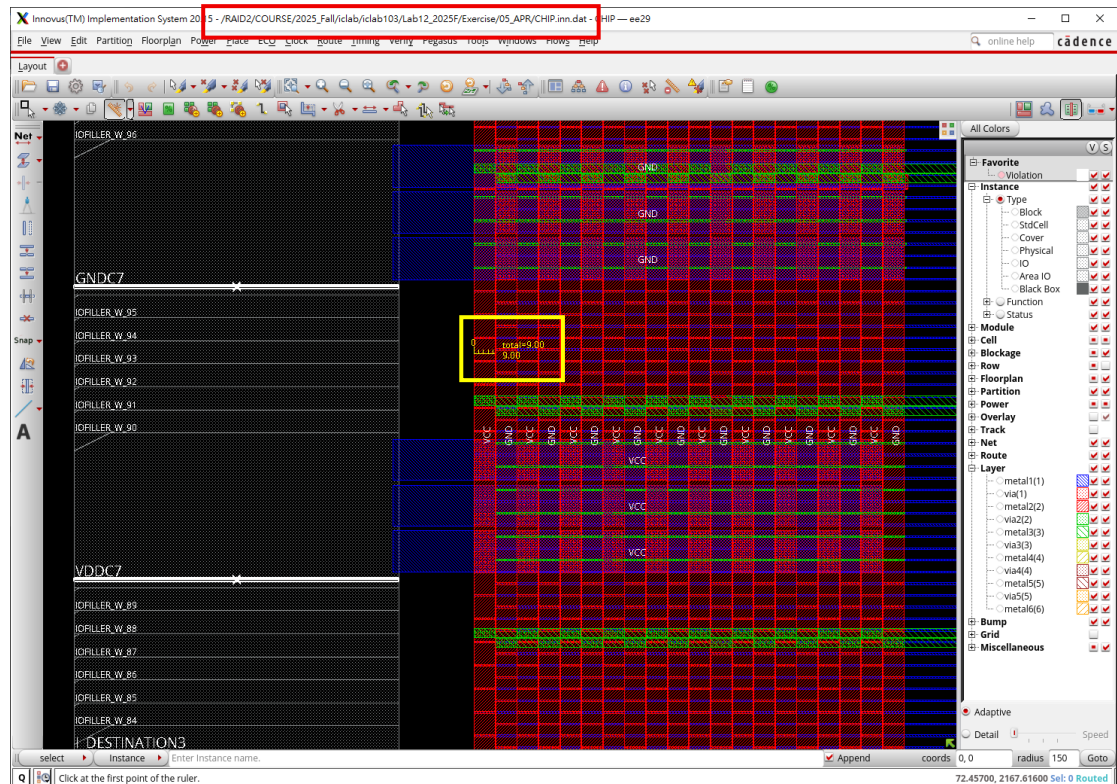
1. Chip Layout View :



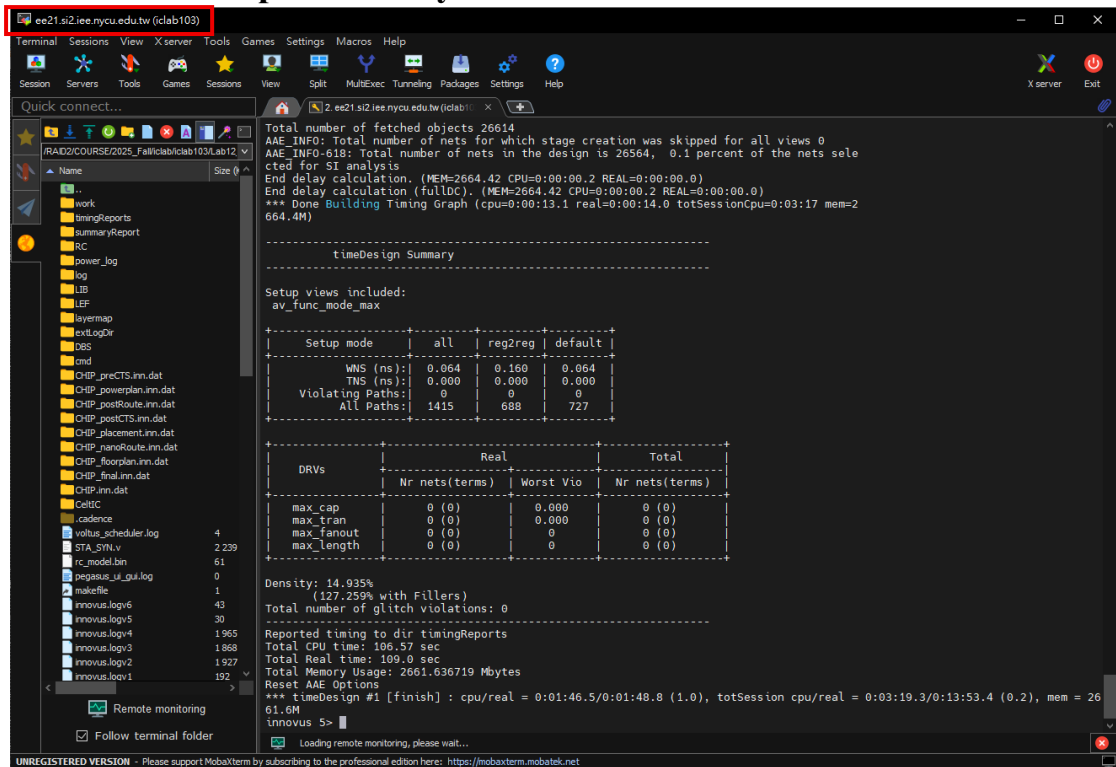
2. Core to IO boundary :



3. Core Ring :



4. Post-Route setup time analysis :



The screenshot shows the output of a post-route setup time analysis in Mobaxterm. The terminal window displays the following information:

```

Total number of fetched objects 26614
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 26564, 0.1 percent of the nets sele
cted for SI analysis
End delay calculation. (MEM=2664.42 CPU=0:00:00.2 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=2664.42 CPU=0:00:00.2 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:13.1 real=0:00:14.0 totSessionCpu=0:03:17 mem=2
664.4M)

-----
timeDesign Summary
-----

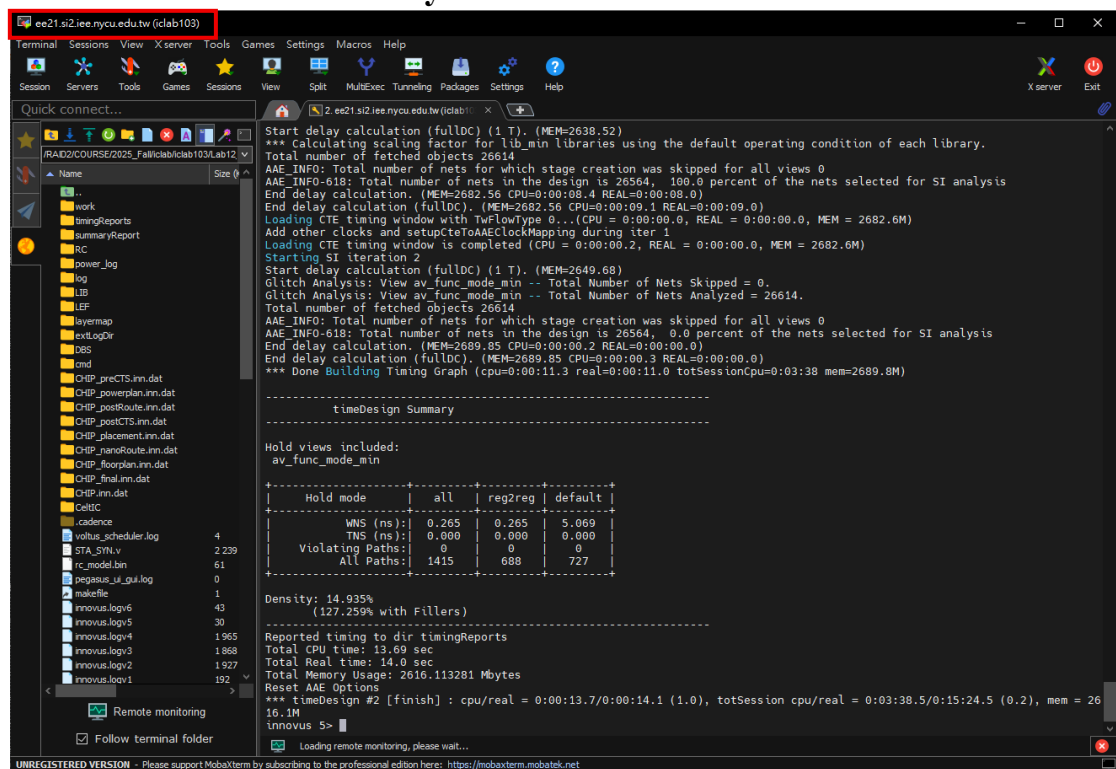
Setup views included:
av_func_mode_max

+-----+-----+-----+
| Setup mode | all | reg2reg | default |
+-----+-----+-----+
| WNS (ns): | 0.064 | 0.160 | 0.064 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 1415 | 688 | 727 |
+-----+-----+-----+

+-----+-----+-----+
| DRVs | | Real | | Total |
+-----+-----+-----+
| | | Nr nets (terms) | Worst Vio | Nr nets (terms) |
+-----+-----+-----+
| max_cap | | 0 (0) | 0.000 | 0 (0) |
| max_fanout | | 0 (0) | 0.000 | 0 (0) |
| max_length | | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+

Density: 14.935%
(127.259% with Fillers)
Total number of glitch violations: 0
-----
Reported timing to dir timingReports
Total CPU time: 106.57 sec
Total Real time: 109.0 sec
Total Memory Usage: 2661.636719 Mbytes
Reset AAE Options
*** timeDesign #1 [finish] : cpu/real = 0:01:46.5/0:01:48.8 (1.0), totSession cpu/real = 0:03:19.3/0:13:53.4 (0.2), mem = 26
61.6M
innovus 5>
  
```

5. Post-Route hold time analysis :



The screenshot shows the output of a post-route hold time analysis in Mobaxterm. The terminal window displays the following information:

```

Start delay calculation (fullDC) (1 T). (MEM=2638.52)
*** calculating scaling factor for lib_min libraries using the default operating condition of each library.
Total number of fetched objects 26614
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 26564, 100.0 percent of the nets selected for SI analysis
End delay calculation. (MEM=2682.56 CPU=0:00:08.4 REAL=0:00:08.0)
End delay calculation (fullDC). (MEM=2682.56 CPU=0:00:09.1 REAL=0:00:09.0)
Loading CTE timing window with TnfLowtype 0... (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 2682.6M)
Add other clocks and setupCtoToAAEClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.2, REAL = 0:00:00.0, MEM = 2682.6M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=2649.68)
Glitch Analysis: View av_func_mode_min -- Total Number of Nets Skipped = 0.
Glitch Analysis: View av_func_mode_min -- Total Number of Nets Analyzed = 26614.
Total number of fetched objects 26614
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 26564, 0.0 percent of the nets selected for SI analysis
End delay calculation. (MEM=2689.85 CPU=0:00:00.2 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=2689.85 CPU=0:00:00.3 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:11.3 real=0:00:11.0 totSessionCpu=0:03:38 mem=2689.8M)

-----
timeDesign Summary
-----

Hold views included:
av_func_mode_min

+-----+-----+-----+
| Hold mode | all | reg2reg | default |
+-----+-----+-----+
| WNS (ns): | 0.265 | 0.265 | 5.069 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 1415 | 688 | 727 |
+-----+-----+-----+

Density: 14.935%
(127.259% with Fillers)
Reported timing to dir timingReports
Total CPU time: 13.69 sec
Total Real time: 14.0 sec
Total Memory Usage: 2616.113281 Mbytes
Reset AAE Options
*** timeDesign #2 [finish] : cpu/real = 0:00:13.7/0:00:14.1 (1.0), totSession cpu/real = 0:03:38.5/0:15:24.5 (0.2), mem = 26
16.1M
innovus 5>
  
```

6. DRC result :

```
ee21.siz.tee.nycu.edu.tw (clab103)
Terminal Sessions View Xserver Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
RAD2/COURSE/2025_Fall/clab103/Lab12
Name Size
work
timingReports
summaryReport
RC
power_log
log
LIB
LEF
layermap
extLogDir
DBS
cmd
CHIP_preCTS.inn.dat
CHIP_powerplan.inn.dat
CHIP_postRoute.inn.dat
CHIP_postCTS.inn.dat
CHIP_placement.inn.dat
CHIP_nanoRoute.inn.dat
CHIP_floorplan.inn.dat
CHIP_final.inn.dat
CHIP.inn.dat
Cadence
cadence
voltage_scheduler.log 4
STA_SIN.v 2 239
rc_model.bin 61
pegasus_ui_gui.log 0
makefile 1
innovus.logv6 43
innovus.logv5 30
innovus.logv4 1965
innovus.logv3 1868
innovus.logv2 1927
innovus.logv1 192
Remote monitoring
Follow terminal folder
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VERIFY DRC ..... Sub-Area: {253.440 2246.400 506.880 2496.000} 101 of 121
VERIFY DRC ..... Sub-Area: 101 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {506.880 2246.400 760.320 2496.000} 102 of 121
VERIFY DRC ..... Sub-Area: 102 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {760.320 2246.400 1013.760 2496.000} 103 of 121
VERIFY DRC ..... Sub-Area: 103 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1013.760 2246.400 1267.200 2496.000} 104 of 121
VERIFY DRC ..... Sub-Area: 104 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1267.200 2246.400 1520.640 2496.000} 105 of 121
VERIFY DRC ..... Sub-Area: 105 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1520.640 2246.400 1774.080 2496.000} 106 of 121
VERIFY DRC ..... Sub-Area: 106 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1774.080 2246.400 2027.520 2496.000} 107 of 121
VERIFY DRC ..... Sub-Area: 107 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2027.520 2246.400 2280.960 2496.000} 108 of 121
VERIFY DRC ..... Sub-Area: 108 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2280.960 2246.400 2534.400 2496.000} 109 of 121
VERIFY DRC ..... Sub-Area: 109 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2534.400 2246.400 2752.800 2496.000} 110 of 121
VERIFY DRC ..... Sub-Area: 110 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 2496.000 253.440 2741.880} 111 of 121
VERIFY DRC ..... Sub-Area: 111 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {253.440 2496.000 506.880 2741.880} 112 of 121
VERIFY DRC ..... Sub-Area: 112 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {506.880 2496.000 760.320 2741.880} 113 of 121
VERIFY DRC ..... Sub-Area: 113 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {760.320 2496.000 1013.760 2741.880} 114 of 121
VERIFY DRC ..... Sub-Area: 114 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1013.760 2496.000 1267.200 2741.880} 115 of 121
VERIFY DRC ..... Sub-Area: 115 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1267.200 2496.000 1520.640 2741.880} 116 of 121
VERIFY DRC ..... Sub-Area: 116 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1520.640 2496.000 1774.080 2741.880} 117 of 121
VERIFY DRC ..... Sub-Area: 117 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1774.080 2496.000 2027.520 2741.880} 118 of 121
VERIFY DRC ..... Sub-Area: 118 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2027.520 2496.000 2280.960 2741.880} 119 of 121
VERIFY DRC ..... Sub-Area: 119 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2280.960 2496.000 2534.400 2741.880} 120 of 121
VERIFY DRC ..... Sub-Area: 120 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2534.400 2496.000 2752.800 2741.880} 121 of 121
VERIFY DRC ..... Sub-Area: 121 complete 0 Viols.
Verification Complete : 0 Viols.
*** End Verify DRC (CPU: 0:00:04.8 ELAPSED TIME: 5.00 MEM: 46.0M) ***
innovus >
Loading remote monitoring, please wait...
```

7. LVS result :

```
ee21.siz.tee.nycu.edu.tw (clab103)
Terminal Sessions View Xserver Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
RAD2/COURSE/2025_Fall/clab103/Lab12
Name Size
work
timingReports
summaryReport
RC
power_log
log
LIB
LEF
layermap
extLogDir
DBS
cmd
CHIP_preCTS.inn.dat
CHIP_powerplan.inn.dat
CHIP_postRoute.inn.dat
CHIP_postCTS.inn.dat
CHIP_placement.inn.dat
CHIP_nanoRoute.inn.dat
CHIP_floorplan.inn.dat
CHIP_final.inn.dat
CHIP.inn.dat
Cadence
cadence
voltage_scheduler.log 4
STA_SIN.v 2 239
rc_model.bin 61
pegasus_ui_gui.log 0
makefile 1
innovus.logv6 43
innovus.logv5 30
innovus.logv4 1965
innovus.logv3 1868
innovus.logv2 1927
innovus.logv1 192
Remote monitoring
Follow terminal folder
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VERIFY DRC ..... Sub-Area: 114 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1013.760 2496.000 1267.200 2741.880} 115 of 121
VERIFY DRC ..... Sub-Area: 115 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1267.200 2496.000 1520.640 2741.880} 116 of 121
VERIFY DRC ..... Sub-Area: 116 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1520.640 2496.000 1774.080 2741.880} 117 of 121
VERIFY DRC ..... Sub-Area: 117 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1774.080 2496.000 2027.520 2741.880} 118 of 121
VERIFY DRC ..... Sub-Area: 118 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2027.520 2496.000 2280.960 2741.880} 119 of 121
VERIFY DRC ..... Sub-Area: 119 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2280.960 2496.000 2534.400 2741.880} 120 of 121
VERIFY DRC ..... Sub-Area: 120 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2534.400 2496.000 2752.800 2741.880} 121 of 121
VERIFY DRC ..... Sub-Area: 121 complete 0 Viols.
Verification Complete : 0 Viols.
*** End Verify DRC (CPU: 0:00:04.8 ELAPSED TIME: 5.00 MEM: 46.0M) ***
innovus > VERIFY_CONNECTIVITY use new engine.
***** Start: VERIFY CONNECTIVITY *****
Start Time: Thu Dec 4 14:03:55 2025
Design Name: CHIP
Database Units: 1000
Design Boundary: (0.0000, 0.0000) (2752.8000, 2741.8800)
Error Limit = 1000; Warning Limit = 50
Check all nets
**** 14:03:55 **** Processed 5000 nets.
**** 14:03:55 **** Processed 10000 nets.
**** 14:03:55 **** Processed 15000 nets.
**** 14:03:56 **** Processed 20000 nets.
**** 14:03:56 **** Processed 25000 nets.
Begin Summary
Found no problems or warnings.
End Summary
End Time: Thu Dec 4 14:03:56 2025
Time Elapsed: 0:00:01.0
***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:01.3 MEM: 33.223M)
innovus >
```

8. Post Layout simulation result :

```
*****
Congratulations!
execution cycles = 50000
clock period = 11.800900ns
*****
$finish called from title "PATIEN.V", line 118.
$finish at simulation time 1003228500
VCS Simulation Report
Time: 1003228500 ps
CPU Time: 22.059 seconds; Data structure size: 8.1Mb
Thu Dec 4 14:05:12 2025
CPU time: 4.452 seconds to compile + 1.761 seconds to elab + 1.096 seconds to link + 22.137 seconds in simulation
14:05 iclab103@ee21:~$ cd ~/Lab12_2025/Exercise/06_POST.js
Loading remote monitoring, please wait...
```

9. Power result :

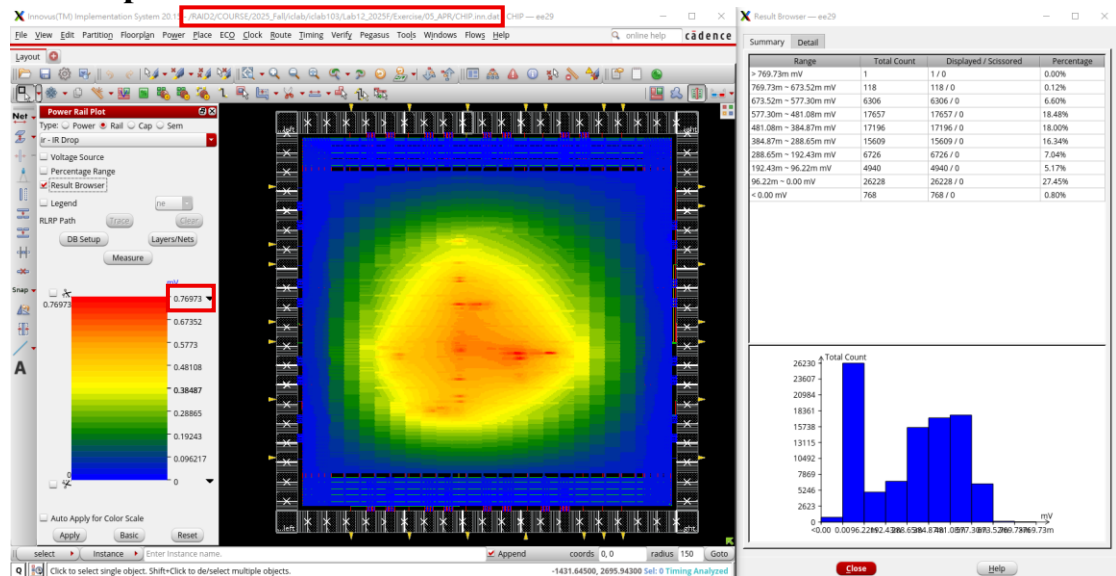
```
** WARN: (VOLTUS_POWR-2152): Instance VDDP7 (VCC3I0D) has no static power.
** WARN: (VOLTUS_POWR-2152): Instance GNDP7 (GNDI0D) has no static power.
** WARN: (VOLTUS_POWR-2152): Instance VDDC0 (VCC0D) has no static power.
** WARN: (VOLTUS_POWR-2152): Instance GND00 (GN0KD) has no static power.
** WARN: (VOLTUS_POWR-2152): Instance VDDC1 (VCC0D) has no static power.
** WARN: (VOLTUS_POWR-2152): Instance GND01 (GN0KD) has no static power.
** WARN: (EMS-27): Message (VOLTUS_POWR-2152) has exceeded the current message display limit of 20.
To increase the message display limit, refer to the product command reference manual.

Total Power
-----
Total Internal Power: 3.78583428 55.0985%
Total Switching Power: 3.06182748 44.5614%
Total Leakage Power: 0.02336695 0.3401%
Total Power: 6.87102871

** WARN: (VOLTUS_POWR-2841): There are some instances in the design which are not connected to any power or ground nets.
These instances will be added to default power/ground rail uti files.
Use 'itaputil list <uti-file>' command to get the list of instances.

** WARN: (VOLTUS_POWR-3424): Cell TIE1 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWR-3424): Cell BUF8 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWR-3424): Cell BUF6 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWR-3424): Cell OR2B1P has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWR-3424): Cell OR2B1S has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWR-3424): Cell A0112 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWR-3424): Cell A0112HS has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWR-3424): Cell MA0222 has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWR-3424): Cell OA13S has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWR-3424): Cell ND2T has no power pin defined in LEF/PGV.
** WARN: (VOLTUS_POWR-3424): Cell AN2S has no power pin defined in LEF/PGV.
```

10. IR Drop Results :



The maximum IR drop : **0.76973 mV**

Strategies to reduce IR Drop issue :

1. Distribute IO and Core PADs (VCC & GND) more evenly.
(IR drop reduced from 1.02mV to 0.76973mV.)
2. Increase the spacing between PADs. (Remove CHIP.io endspace = 31.)
3. Maintain 19 PADs on each side. (Let the chip shape remains square.)
4. Allocate 4 Core Power PADs and 2 IO Power PADs on each side.